

National Institute of Technology Kurukshetra
[B.Tech. (EE-Branch) II- Semester
Mid-Semester-Exam-II

[Advanced Engineering Physics (AEP); Course No. PHIC104]

Note: All questions are compulsory

Time allotted: 50 min.

Max. Marks: 15

1.	Starting from Maxwell's equations, derive the expression for the velocity of electromagnetic waves in free space and show that: $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$	[5]
2.	Derive the following conditions: (a) Tangential component of electric field is continuous (b) Normal component of magnetic flux density	[3]
3.	A toroid has 800 turns and carries a current of 3 A. The mean radius is 0.25 m. Using Ampere's Circuital Law, find the magnetic field inside the toroid.	[2]
4.	Why are ground waves more effective at low frequencies? Explain how the conductivity of the Earth affects their propagation.	[2]
5.	Prove that susceptibility of superconductor is -1 and relative permeability is zero. List any five applications of superconductivity in practical fields.	[3]